



MA 588: MATHEMATICAL CONTROL THEORY

Course Code	MA 588
Course Name	MATHEMATICAL CONTROL THEORY
L-T-P-C	3-1-0-4
Prerequisites	MA511, M513 & MA521
Intended for	M.Sc. / BTech 3rd and 4th years / MTech/ PhD (All Branches)
Approval	56th BoA (Jul 2024)

Course Contents

- **Module I:** Solution of Uncontrolled Systems: - Spectral Form, Exponential Matrix, Repeated Roots, Solution of Controlled System – State Space, Control Space, Time Varying Systems, Discrete Time Systems. [8 Lectures]
- **Module II:** Linear Control System, State Transition Matrix, Properties of State Transition Matrix, Controllability, Kalman Matrix, Kalman Condition for Controllability, Controllability Matrix and Related Theorems. [10 Lectures]
- **Module III:** Semigroup of Linear Operators, Infinitesimal Generators, Strongly Continuous and Uniformly Continuous Semigroups, Properties of Semigroups, Hille-Yosida Theorem, Compactness and Differentiability of semigroups, Analytic Semigroup, Semigroup of Compact Operators, Abstract Differential Equations and Their Solutions. [12 Lectures]
- **Module VI:** Solution of Infinite Dimensional Control Systems, Controllability - Exact Controllability, Approximate Controllability, Null Controllability, Controllability Map, Controllability Grammian. Necessary and Sufficient Conditions for Exact and Approximate Controllability.

Text Books

1. R. F. Curtain and Hans Zwart, **An Introduction to Infinite-Dimensional Linear System Theory**, Springer, 1995.
2. S. Barnett, **Introduction to Mathematical Control Theory**, Clarendon Press, 1985.

References

1. A. Pazy, **Semigroup of Linear operators and Applications to Partial Differential Equations**, Springer Verlag, 1983
2. D. E. Kirk, **Optimal Control Theory**, Dover Publications, 2004.
3. W. L. Brogan, **Modern Control Theory, Third Edition**, Prentice Hall, 1991.